Lecture 13: The structural / empirical IO research process

ECON 7510

Cornell University

Adam Harris

Outline

1. Research questions
2. Data
3. Developing a model
4. Presenting and writing

Research questions

Where to begin

Approaches to identifying research questions

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   - → What *new* questions can I answer with this data?
   - → What old questions can I *better* answer with this data?

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- **Interesting to you**: If you're not really interested in it, you're setting yourself up for years of tedious work.

Data

Where to look for data

1. Government
2. Standard non-government data sets
3. Firms

Government data

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- Regulatory agencies. For example,
  - Occupational Safety and Health Administration
  - Centers for Medicare & Medicaid Services
  - Federal Communications Commission (e.g., broadband auctions)
  - State departments of transportation
  - Federal Motor Carrier Safety Administration
  - → If they don't want to share data, FOIA may be an option.

Standard non-government data sets

- Compustat—data from reports of publicly-traded firms
- Orbis
- Neilsen scanner data

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- Industry media—trade journals, online publications, podcasts, etc.
- Industry events—you'll be an oddity, but that's okay.

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  - Depending on how things go, might wait until 2nd or 3rd meeting to broach the subject of data.

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- When giving them a proposal or summary of findings, consider style as well as substance.

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Notes:

- DUA is generally *not* an agreement between you and the firm; it's an agreement between Cornell and the firm, which you being an agent of the university.
- Getting the DUA in place can take a while.

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  - → Estimate this model; think through identification argument.
- What element should I incorporate next?
  - → Does incorporating this pose a problem for identification?
  - → Does incorporating this pose a problem for estimation?

Presenting and writing

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- Audience member are human beings with their own lives, which means
  - they have finite cognitive capacity. Don't waste that capacity by overwhelming them with unnecessary details.
  - they have spent maybe 1% as much time thinking about your topic as you have. Think carefully about what knowledge you can and cannot assume.
    - → Especially important when presenting to an audience outside of your field.

Principle 2:

Brevity

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- Aspect ratio 16:9 is much better than the default.
  `\documentclass[11pt,xcolor=table,mathserif,aspectratio=169]{beamer}`

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  - You don’t want to come across as defensive. May even want to acknowledge weaknesses where appropriate.

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- Anticipate and be prepared for questions.
  - Lots of backup slides.
  - However, you needn't try to forestall all potential questions by answering them before they're asked.